

Sohail Mohammad

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SUMMARY

Senior Data Engineer with 5+ years of experience delivering multi-cloud data platforms, real-time analytics, and enterprise AI solutions across various domains. Proven expertise across **AWS, Azure, and GCP**, with a strong focus on performance optimization, cost savings, and regulatory compliance.

- Engineered scalable ETL/ELT pipelines using AWS Glue, Azure Data Factory, and GCP Dataflow, reducing processing latency by 40% (15 minutes → 9 minutes) and boosting throughput by 45% (22K → 32K jobs per hour).
- Built high-throughput real-time analytics pipelines with Spark Streaming, Kafka, and Azure Event Hub to process 10M+ events per minute, reducing decision latency from 3 seconds to under 1 second across mission-critical domains.
- Led cloud migrations to AWS S3, Azure Data Lake, and GCP Storage, delivering 100% data integrity (zero loss during 5TB+ transfers) and ensuring HIPAA/GDPR compliance for sensitive datasets.
- Automated CI/CD pipelines with Jenkins, Azure DevOps, and GitHub Actions, reducing ML deployment cycles by 20% (5 days → 4 days) and ensuring reproducible, cross-environment deployments.
- Designed Medallion architectures and optimized PySpark/SQL workloads, improving dataset processing times by 35% (90 minutes → 58 minutes) while lowering infrastructure costs.

EXPERIENCE

Senior Data Engineer

October 2024 – Present

DTE Energy

Detroit, USA

- Architected a multi-tenant GenAI platform across AWS, Azure, GCP, enabling real-time LLM inference for 1,000+ concurrent users (vs <200 before) with cloud-native microservices and Kubernetes.
- Implemented retrieval-augmented generation (RAG) pipelines and memory frameworks using Databricks, Azure Data Factory, and Delta Lake, delivering query responses in under 2 seconds (vs 6–7 seconds earlier).
- Optimized inference via caching, batching, and tuning, improving throughput and unit cost while meeting strict business SLAs.
- Automated CI/CD, rollback, and test automation for AI-powered services using Jenkins, GitHub Actions, and Docker, reducing release cycles by 30%.
- Designed APIs and SDKs for LLM training and fine-tuning, and integrated agentic frameworks (LangGraph, Semantic Kernel, LlamaIndex) to automate multi-step enterprise workflows.
- Implemented AI guardrails including prompt filtering, safety checks, and red-teaming, along with metadata and lineage pipelines for model traces, ensuring SOC 2 and GDPR compliance.
- Benchmarked foundation models and fine-tuned variants across diverse tasks to guide trade-offs in cost, latency, and quality, establishing platform-wide standards.
- Hardened infrastructure with disaster recovery, achieving Recovery Point Objective (RPO) under 5 minutes and Recovery Time Objective (RTO) under 15 minutes through automated failover.

Senior Data Engineer

Jan 2024 – Sep 2024

T-Mobile

Kansas, USA

- Delivered Azure-based ETL and streaming solutions using Data Factory, Databricks, Event Hub, and Spark Streaming, improving pipeline performance by 40% (processing 1.2B → 1.7B rows/hr) for telecom analytics.
- Designed end-to-end pipelines and Medallion architecture in Databricks (bronze, silver, gold layers) with PySpark, reducing dataset processing times by 35% and improving machine learning feature engineering.
- Developed real-time ingestion pipelines with Event Hub, Azure Stream Analytics, and Spark Streaming, supporting network monitoring and fraud detection in <1s latency (vs 4–5s earlier).
- Migrated 8TB of enterprise and customer datasets to ADLS and Cosmos DB with zero data loss, ensuring GDPR compliance.
- Managed Cosmos DB and Azure SQL Database, improving query performance by 40% through schema optimization, indexing, and partitioning for ML queries.
- Automated CI/CD deployments of Databricks and ML pipelines using Azure DevOps, GitHub Actions, and Terraform, reducing release cycles by 20%.

- Implemented governance and data lineage with Azure Purview, enabling metadata tracking, lineage, and reproducibility for ML experiments.
- Built observability dashboards in Splunk and Azure Monitor to monitor pipeline performance, drift, and failures, reducing troubleshooting time by 30%.
- Collaborated in Agile teams (sprint planning, stand-ups, retrospectives) to deliver data engineering and ML projects on schedule.

Data Engineer

June 2020 – Sep 2023

Citi Bank

Hyderabad, India

- Designed and implemented end-to-end ETL/ELT pipelines on AWS Glue, EMR, Lambda, and S3, enabling scalable ML and generative AI workloads with improved reliability.
- Engineered enterprise data lakes and Amazon Redshift warehouses from heterogeneous sources (Oracle, SQL Server, flat files, mainframe), improving ML feature engineering productivity by 45%.
- Built Medallion-style ETL frameworks in Databricks and Talend (bronze, silver, gold layers), ensuring analytics-ready datasets for downstream ML pipelines.
- Integrated Amazon EKS into CI/CD pipelines with Jenkins, GitLab, and Bitbucket for automated deployment of containerized ML and GenAI applications.
- Automated archival, ingestion, and testing workflows by developing AWS SDK/CLI and Python/Scala scripts, supporting S3 Glacier, REST API data ingestion, and A/B testing of ML models.
- Provisioned and configured Amazon EC2 instances (Linux/Ubuntu AMIs) to accelerate ML model training and inference workloads.
- Optimized PySpark and SQL jobs in EMR and Redshift, reducing data processing time by 30% and lowering cloud infrastructure costs.
- Delivered interactive dashboards and reports with Power BI and Tableau, presenting ML and GenAI insights to stakeholders for data-driven decisions.
- Led full lifecycle data engineering projects (analysis, design, modeling, testing, reporting), ensuring business requirements translated into scalable ML/AI solutions.

Junior Data Engineer

Dec 2019 – May 2020

Citi Bank

Hyderabad, India

- Developed and optimized ETL pipelines using Apache Hive, Sqoop, and SQL, reducing processing latency by 35% for large-scale analytics.
- Built ingestion pipelines with Sqoop to transfer data from Oracle, SQL Server, and MySQL into Hadoop HDFS, ensuring complete data integrity.
- Authored and optimized advanced Hive queries (joins, subqueries, window functions) to process petabyte-scale datasets in Hadoop HDFS.
- Orchestrated batch and streaming workflows across Hive, Sqoop, and Kafka using Jenkins, with reliable, dependency-aware execution.
- Designed star and snowflake schema models in Hive to support enterprise data marts and high-performance reporting.
- Automated ingestion and transformation tasks with Java-based utilities, improving ETL efficiency and reducing manual overhead.
- Implemented CI/CD pipelines via Jenkins to deploy Hive scripts and Java applications, cutting release cycles by 25%.
- Built dashboards by integrating Hive with Tableau, delivering actionable insights to business stakeholders.

EDUCATION

Campbellsville University

July 2025

Master of Science - Computer Science

Osmania University

May 2023

Bachelor of Engineering - Electronics and Communication

SKILLS

Data Engineering & Orchestration: ETL/ELT, Pipeline Orchestration, Lakehouse, Data Marts, Medallion Architecture, Batch & Streaming (Apache Spark, Kafka, Airflow, AWS Glue, Azure Data Factory, GCP Dataflow), Retrieval-Augmented Generation (RAG), Snowflake Schema.

Big Data & Distributed Systems: Hadoop, Spark (PySpark, Spark Streaming), Hive/Impala, Databricks, Amazon Redshift.

Hadoop Distribution: Cloudera Distribution, Hortonworks.

Machine Learning & AI: ML Pipelines, MLflow, Kubeflow, Vertex AI, SageMaker, Model Deployment & Monitoring, A/B Testing, Generative AI (LLM/RAG integration, LangChain, LlamaIndex).

Programming Languages & Scripting: Python, Java, Scala, SQL, R, Shell Scripting; Libraries: Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow, FastAPI, Jupyter.

Web Frameworks: Streamlit, Flask, R Shiny, FastAPI, Spring Boot.

Cloud Platforms & DevOps: AWS, Azure, GCP, Databricks, Kubernetes, Docker, Terraform, Jenkins, GitHub Actions, GitLab CI/CD, Azure DevOps, CodePipeline, Splunk.

Databases: SQL (Oracle, SQL Server, MySQL, PostgreSQL), NoSQL (MongoDB, DynamoDB, Cosmos DB, Cassandra), Data Warehouses (Snowflake, Redshift).

Data Visualization & BI: Tableau, Power BI, Google Analytics, Salesforce Analytics, Excel.

Operating Systems: Linux (Ubuntu, Red Hat), macOS, Microsoft Windows (Server & Desktop).

Governance, Security & Compliance: Azure Purview, Collibra, Data Catalog, Metadata Management, Data Lineage, RBAC/IAM, Encryption, Key Vault, Disaster Recovery, Compliance (HIPAA, GDPR, SOC 2).

Other Tools & Methodologies: Jira, Confluence, Agile, Scrum, Kanban.

CERTIFICATIONS

Microsoft Certified Fabric Data Engineer - Associate (DP-700): [Link](#)

Microsoft Certified Power BI Data Analyst (PL-300): [Link](#)

Microsoft Office Specialist: Microsoft Excel Expert (MO-201): [Link](#)

Big Data Analytics with Hadoop and Apache Spark: [Link](#)